ALICE L WALTON SCHOOL OF MEDICINE

Spectrum of Activity and Indications–4 50S inhibitors

Last updated May 30, 2025 · 5 min read · CASE CHALLENGE

Authors: Dipitman Bose, Tinger Hintze
Editors: Jordan Finn, Russell Johnson

Listen to this Brick
0:00 / 10:23 1x

Learning Objectives (1)

After completing this brick, you will be able to:

1. Describe the spectrum of activity of 50S protein synthesis inhibitors.

CASE CONNECTION

PR is a 46-year-old female who presents to the urgent care clinic with shortness of breath, productive cough, and a temperature of 100.8°F. A chest X-ray reveals a left lower lobe infiltrate. Her past medical history includes back pain and high cholesterol. PR also has a Penicillin allergy. You check PR's history and find that she has a history of QT prolongation. The attending physician and you make a diagnosis of *pneumonia*.

The attending physician asks you to make a recommendation for PR. What is the empiric treatment regimen you would recommend for PR?

GO TO CONCLUSION

The 50S protein synthesis inhibitors disrupt the bacterial translation of proteins and are typically bacteristatic. These include the macrolides (azithromycin, clarithromycin), linezolid, and streptogramins.

What is the spectrum of activity for 50S protein synthesis inhibitors? (LO#1)

Macrolides

Macrolides have excellent coverage for **atypical** (*Chlamydia, Mycoplasma, Mycobacterium*, and *Legionella*). They also have good *S. pneumoniae, Haemophilus, Neisseria*, and *Moraxella* coverage. Given that these pathogens are commonly involved in respiratory infections, this makes the macrolides a treatment option for upper and lower respiratory tract infections.

CLINICAL CORRELATION

Community-acquired pneumonia (CAP) is a respiratory infection acquired from the community as opposed to hospital-acquired pneumonia (HAP). If it is determined that the patient does NOT require hospitalization, empiric regimens are designed to cover common pathogens: *Streptococcus pneumoniae, Haemophilus influenzae*, and atypical pathogens (ie, *Mycoplasma pneumoniae, Legionella pneumophila*, and *Chlamydia pneumoniae*).

The most common first-line treatment option is a beta-lactam (usually a third generation cephalosporin like cefdinir, or amoxicillin/ clavulanate) AND a macrolide (most often azithromycin for the atypical coverage). Doxycycline can be used in the place of the macrolide if there are contraindications.

A respiratory fluroquinolone (monotherapy) can also be used for CAP.

Clindamycin

Clindamycin is used against **anaerobic infections above the diaphragm** and aerobic Gram-positive cocci. It is not effective against enterococcus and gram-negative pathogens including Gram-negative anaerobic bacteria.

Linezolid and Streptogramin

The coverage for these agents is similar to Vancomycin (gram-positive coverage). Linezolid covers **gram positive bacteria**. More importantly, they cover *Staphylococci* (MRSA) and Vancomycin resistant *Enterococci (VRE)*. They are administered orally and can be used in an outpatient setting.

Streptogramins (Quinupristin/Dalfopristin) also have excellent **gram-positive coverage** including M

What types of bacteria do Linezolid and Streptogramins cover?

CLINICAL REASONING

Why is Ciprofloxacin not called a respiratory fluoroquinolone? Ciprofloxacin is not considered a respiratory fluoroquinolone primarily because it has poor activity against *Streptococcus pneumoniae*, which is a common pathogen in community-acquired *pneumonia (CAP)*.

Treating Tick-borne Infections

Ticks, fleas, and lice are hosts to atypical bacteria that can potentially cause life-threatening diseases. In our [atypical pneumonia brick](), we have discussed the pathogenesis of some of these diseases and their clinical presentations. From a treatment point of view, the first-line treatment in most of these cases is doxycycline. The table 1 below summarizes the infections caused by atypical bacteria, the organism, and the first-line treatment for the diseases.

Table 1: Treatment of Fastidious pathogens

DISEASE	ORGANISM	TREATMENT
Rocky Mountain Spotted Fever	*Rickettsia rickettsii*	Doxycycline
Lyme Disease	*B. burgdorferi*	Doxycycline
Ehrlichiosis	*Ehrlichia chaffeensis*	Doxycycline
Legionnaires' disease	*Legionella*	Azithromycin or Levofloxacin

What is the first-line treatment for infections caused by atypical bacteria transmitted by ticks, fleas, and lice?

CASE CONNECTION BACK TO INTRODUCTION

Based on your reading, what is the empiric treatment regimen you would recommend for PR?

PR is diagnosed with pneumonia, which is most likely community-acquired. If it is determined that the patient does not require hospitalization, empiric regiments are designed to cover common pathogens causing CAP. These include *Streptococcus pneumoniae, Haemophilus influenzae*, and atypical pathogens (ie, *Mycoplasma pneumoniae, Legionella pneumophila*, and *Chlamydia pneumoniae*). She is otherwise healthy. Your options are that you can treat her with Amoxicillin/Clavulanate, Doxycycline, or a macrolide. However, she has a penicillin allergy; amoxicillin/clavulanate would not be an appropriate choice. Also, macrolides increase the risk of QT prolongation. Hence, in this case, you recommend a regimen of doxycycline for PR. You ask her to keep recording her temperature and her cough. Doxycycline use can cause photosensitive reactions. You can advise PR to appropriately protect herself from the sun.

Summary

- Macrolides, such as azithromycin and clarithromycin, have excellent coverage for atypicals like Chlamydia, Mycoplasma, and Legionella, and are used for respiratory infections.
- Neisseria is not typically covered by macrolides, despite their good coverage for other atypicals and S. pneumoniae.
- Clindamycin is effective against anaerobic infections above the diaphragm and aerobic gram.
- Linezolid and Streptogramins provide excellent gram-positive coverage, including MRSA, with Linezolid also covering VRE.
- The first-line treatment for most tickborne infections is doxycycline.

Review Questions

1. Which of the following types of bacteria are effectively treated with clindamycin?

A. Aerobic gram-positive cocci
B. Anaerobic bacteria below the diaphragm
C. Aerobic gram-negative bacilli
D. *Enterococcus*

Hide Explanation

The correct answer is aerobic gram-positive cocci (A). Clindamycin is effective in covering the aerobic gram-positive cocci and anaerobic bacteria above the diaphragm. Choices (B), anaerobic bacteria below the diaphragm, and aerobic gram-negative bacilli (C) are not covered by clindamycin. Clindamycin has minimal *Enterococcus* (D) activity.

2. The first-line treat

A. Linezolid
B. Azithromycin
C. Cephalexin
D. Doxycycline

Hide Explanation

The correct answer is doxycycline (D). For many diseases caused by atypical bacteria, such as Lyme and Rocky mountain spotted fever Doxycycline is the first line treatment. Linezolid (A) is incorrect because it does not cover the atypicals. Azithromycin (B) is incorrect because macrolides do not cover any of the atypicals. Cephalexin (C) is incorrect because cephalosporins do not cover any of the atypical bacteria.

3. Which of the following can treat a VRE infection?

A. Azithromycin
B. Linezolid
C. Amoxicillin
D. Aztreonam

D. Aztreonam

Hide Explanation

The correct answer is Linezolid (B). Linezolid covers Gram-positive bacteria including MRSA and VRE. Azithromycin (A) are most effective against atypicals. Amoxicillin (C) is not effective against VRE. Aztreonam (D) covers many Gram-negative bacteria including *Pseudomonas*.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Describe the spectrum of activity of 50S protein synthesis inhibitors.**

Objective Confidence: 1/1